

AI TOOLS FOR RESEARCH

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Background

WHERE WE ARE IN 2026

AI Is No Longer Just a Tool

AI is moving from prediction and generation toward reasoning, planning and action.



Think for a moment: the AI tool you used most recently — did it only answer your question, or did it actually perform a sequence of tasks for you?

THE AI LANDSCAPE IN 2026

Seven Forces Shaping AI in 2026

Not isolated technologies — interconnected developments that reinforce one another.



1

Multimodal foundation models

See, hear, read, respond

2

Reasoning-oriented models

Structured problem solving

3

Agentic AI

Plans, uses tools, acts

4

Small & efficient models

On-device, affordable

5

AI for science & engineering

A research collaborator

6

Physical & embodied AI

Robots, drones, mobility

7

Trustworthy & sovereign AI

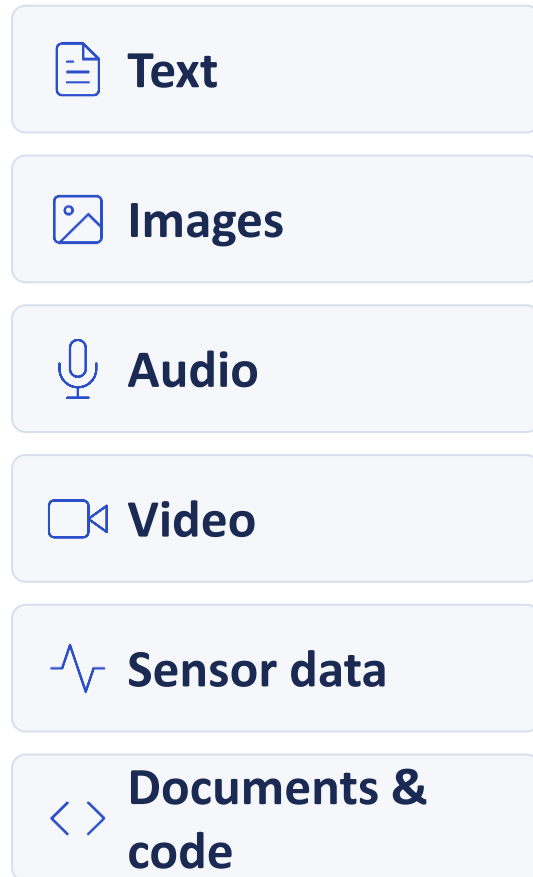
Safe, fair, accountable

FORCE 01 · MULTIMODAL FOUNDATION MODELS

AI That Can See, Hear, Read and Respond

Real-world intelligence is naturally multimodal — one model, many kinds of input.

INPUTS



WHAT IT ENABLES



FORCE 02 · REASONING-ORIENTED MODELS

From Pattern Recognition to Structured Problem Solving

New models spend more time 'thinking' — decomposing, checking and revising before answering.

TRADITIONAL AI

“Recognise and predict”

One fast pass from input to output — great for perception, weaker on multi-step problems.

REASONING-ORIENTED AI

“Break down, analyse, verify, solve”

Multi-step solving · planning · maths & coding · self-evaluation · tool-supported reasoning



A caution: reasoning models can produce convincing but wrong chains of thought. Reasoning does not eliminate hallucination — verification, external tools and human judgment remain essential.

FORCE 03 · AGENTIC AI

AI That Can Plan and Act

An agent pursues a goal through multiple coordinated actions — with a human in the loop.



Chatbot responds to a prompt, one turn at a time



AI agent pursues a goal through many coordinated actions

WHERE AGENTS ARE ALREADY WORKING



Research assistance



Software development



Business workflows



Cybersecurity operations



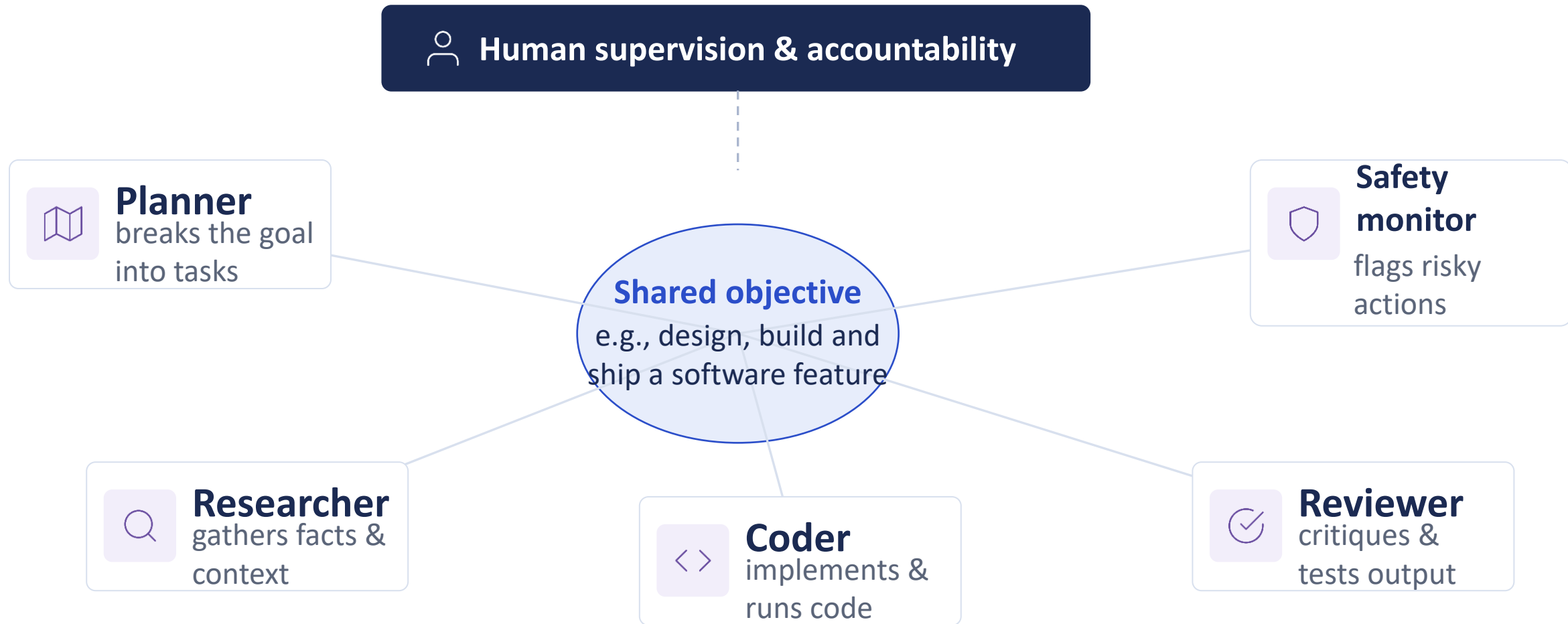
Personalised education



Healthcare administration

When AI Systems Work as Teams

“Specialisation improves capability, but coordination determines reliability.”



Coordination challenges: conflicting outputs · communication overhead · error propagation · security · unclear accountability

FORCE 04 · SMALL & EFFICIENT MODELS

Bigger Is Not Always Better

The right question is not 'how large?' but 'how appropriate for the task, budget and place?'

 Large general model (cloud)		 Small specialised model (edge / device)
Pay-per-use; grows with scale	Cost	Low, predictable, often one-time
Network latency on every call	Speed	Millisecond, local responses
Data leaves the device	Privacy	Data can stay on the device
Data-centre scale power	Energy	Frugal enough for phones & sensors
Limited, via prompts / APIs	Customisation	Deep fine-tuning for your domain
Needs connectivity	Offline use	Works in rural & field settings

How models get small: compression · quantisation · distillation · parameter-efficient fine-tuning (e.g., LoRA) · edge AI · domain-specific language models

Why it matters for India: affordable, multilingual, on-device AI for classrooms, clinics, farms and low-connectivity regions.

AI Is Becoming a Scientific Collaborator

Not just generating text — discovering patterns, proposing hypotheses and accelerating experiments.



AI in the loop of discovery

*observe → model →
hypothesise → experiment*



Drug & molecule discovery

protein structure, candidate screening



Climate & weather

faster, finer-grained forecasting



Material discovery

batteries, catalysts, semiconductors



Space & remote sensing

earth observation at scale



Healthcare

imaging, triage, drug repurposing



Agriculture

crop health, soil, yield advisories



Engineering design

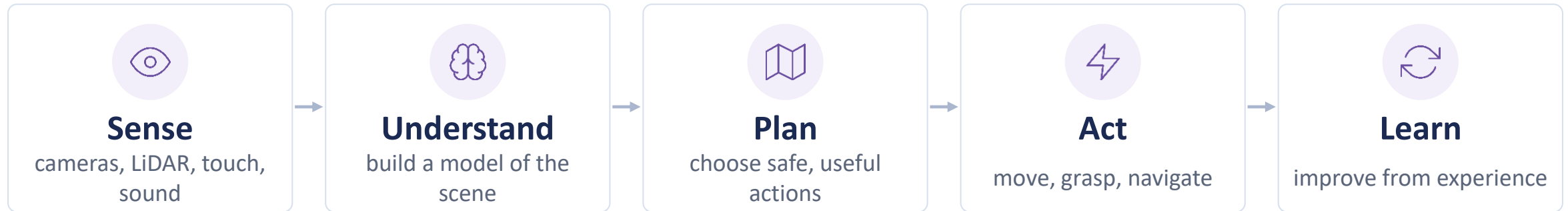
simulation, optimisation, digital twins

The valuable combination: deep domain knowledge + AI expertise. Landmark example: AI-based protein-structure prediction was recognised with the 2024 Nobel Prize in Chemistry.

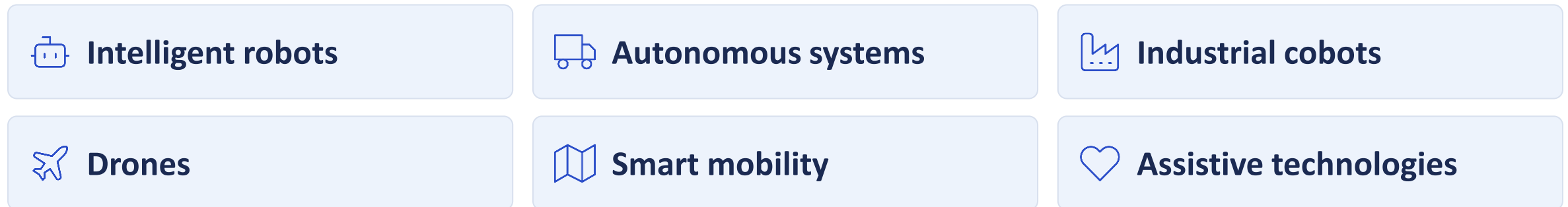
FORCE 06 · PHYSICAL & EMBODIED AI

When Intelligence Enters the Physical World

Robots, vehicles and machines that must understand uncertainty, space, time — and safety.



WHERE EMBODIED AI IS ARRIVING



A wrong chatbot answer is inconvenient. A wrong robotic action can be dangerous. Safety, testing and verification are central research challenges here.

Software Is Becoming AI-Assisted and AI-Defended

The same capability strengthens both builders and defenders — and attackers.

<> AI for software engineering

-  Code generation & pair programming
-  Test generation and coverage
-  Debugging and root-cause analysis
-  Documentation from code
-  Understanding large legacy systems

🛡️ AI for cybersecurity

-  Threat detection & anomaly hunting
-  Behaviour analysis across networks
-  Phishing & fraud detection
-  Faster incident response
-  Vulnerability assessment at scale

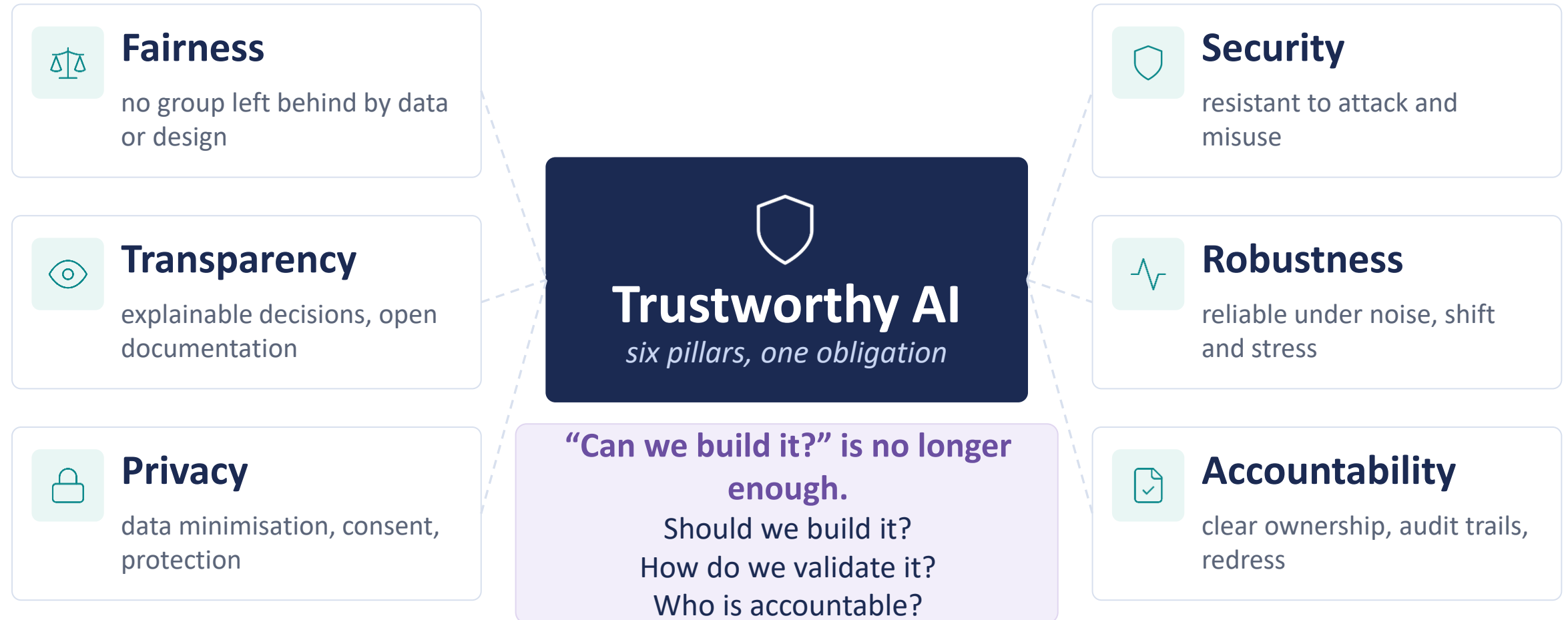


AI can strengthen both defenders and attackers. Generated code must still be tested for correctness, security, bias and licensing — fundamentals are non-negotiable.

FORCE 07 · TRUSTWORTHY & SOVEREIGN AI

Powerful AI Must Also Be Trustworthy

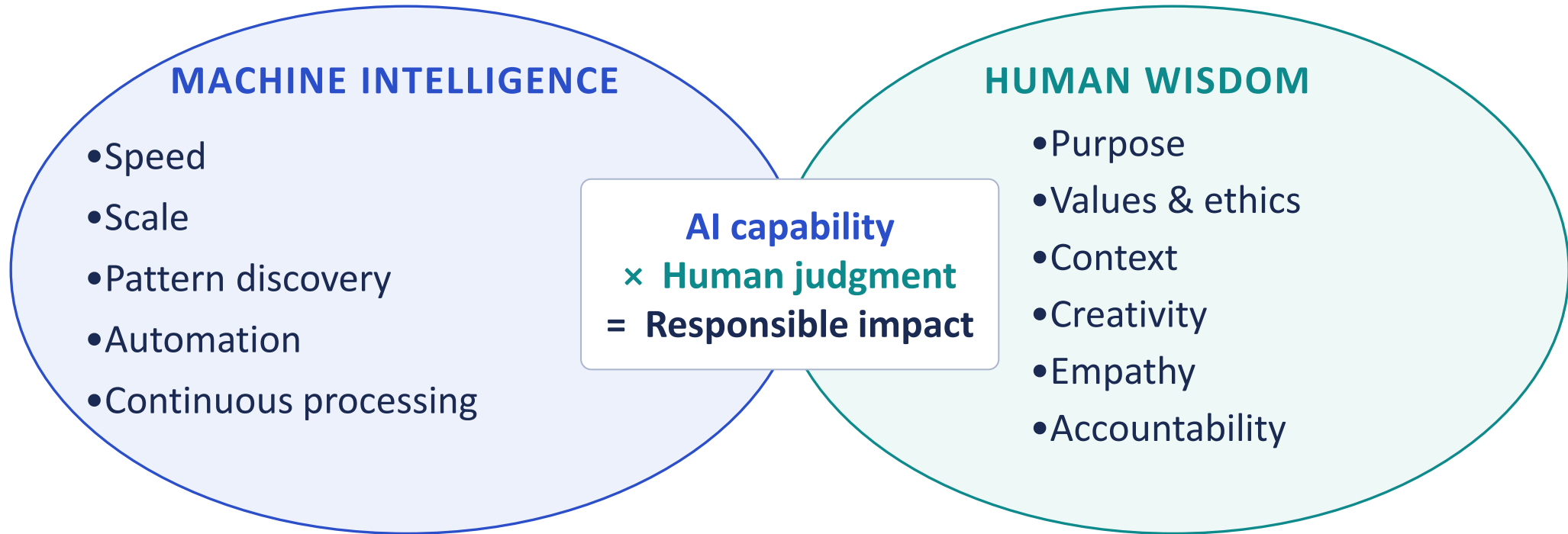
Capability without trust will not be adopted — and should not be.



THE PARTNERSHIP

The Future Is Not Human Versus AI

It is human purpose, amplified by machine capability.



Tell me: what is one human capability that you would never want an AI system to replace?

YOUR ROADMAP

From AI User to AI Creator

Five stages — each one builds on the previous; none can be skipped.

Keep evidence:

*GitHub repositories · experiment logs
proper documentation · honest evaluation*

“Do not build ten copied projects. Build two projects that solve real problems — and can withstand questioning.”

1 • Foundations

Mathematics, algorithms, programming, databases, operating systems



2 • AI Core

Machine learning, deep learning, NLP, computer vision, optimisation



3 • Modern AI

Transformers, generative AI, multimodal models, agents, retrieval systems



4 • Responsible deployment

MLOps, evaluation, security, privacy, explainability, governance



5 • Evidence of ability

Projects, papers, open-source contributions, internships, competitions

CLOSING MESSAGE

The future of AI is also a choice

“AI will shape the future, but informed, ethical and courageous people will decide what kind of future it becomes.”

LEARN DEEPLY

BUILD RESPONSIBLY

LEAD WITH PURPOSE



Learn → Create → Research → Serve → Lead